University of Toronto
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Part 1: Queries

1. Frugal doctors

– prices of generic drugs.

$GenericPrices(DIN, brand, price) := \Pi_{DIN,brand,price}(Generic \bowtie Price)$

– prices of product drugs.

$ProductPrices(DIN) := \Pi_{DIN}(Product \bowtie Price)$

– prices of drugs in prescriptions.

$PrescriptionPrices(RxID, date, patient, drug, doctor, dosage, note) :=$
$\Pi_{RxID,date,patient,drug,doctor,dosage,note}(Prescription \bowtie_{drug=DIN} Price)$

– cheapest generic alternatives of a product.

$CheapestGeneric(DIN) := \Pi_{DIN}(GenericPrices -$
$\Pi_{d1.DIN,d1.brand,d1.price}(\sigma_{\substack{d1.brand=d2.brand \\ \wedge \\ d1.DIN \neq d2.DIN \\ \wedge \\ d1.price > d2.price}}(\rho_{d1}GenericPrices \times \rho_{d2}GenericPrices)))$

– products with no generics.

$NoGenerics(DIN) := ProductPrices - \Pi_{brand}Generic$

– doctors who have prescribed two different drugs.

$TwoDrugs(doctor, drug) :=$
$\Pi_{p1.doctor,p1.drug}(\sigma_{\substack{p1.doctor=p2.doctor \\ \wedge \\ p1.RxID \neq p2.RxID \\ \wedge \\ p1.drug \neq p2.drug}}\rho_{p1}PrescriptionPrices \times \rho_{p2}PrescriptionPrices)$

– doctors who have prescribed drugs only with no generics or the cheapest generics.

$FrugalDoctors(doctor) := \Pi_{doctor}(TwoDrugs - \Pi_{doctor,drug}(TwoDrugs \bowtie_{drug=DIN}$
$((\Pi_{DIN}Product \cup \Pi_{DIN}Generic) - (CheapestGeneric \cup NoGenerics))))$

2. Potential doctor shopping

– patients and the drug they have been prescribed.

$PatientDrug(OHIP, name, phone, drug, doctor) :=$
$\Pi_{OHIP,name,phone,drug,doctor}(Patient \bowtie_{patient=OHIP} Prescription)$

– patients who have been prescribed drugs from different doctors.

$DiffDocs(OHIP, name, phone, drug1, drug2) :=$

$$\Pi_{p1.OHIP,p1.name,p1.phone,p1.drug,p2.drug}\sigma_{\substack{p1.OHIP=p2.OHIP\\ \wedge\\ p1.doctor\neq p2.doctor}}(\rho_{p1}PatientDrug\times\rho_{p2}PatientDrug)$$

– patients who have been prescribed drugs with the same DIN from different doctors.

$SameDIN(OHIP, name, phone) := \Pi_{OHIP,name,phone}\sigma_{drug1=drug2}DiffDocs$

– patients who have been prescribed equivalent generic drugs from different doctors.

$SameBrand(OHIP, name, phone) := \Pi_{OHIP,name,phone}\sigma_{g1.brand=g2.brand}(DiffDocs \bowtie_{drug1=g1.DIN} \Pi_{g1.DIN,g1.brand}\rho_{g1}Generic \bowtie_{drug2=g2.DIN} \Pi_{g2.DIN,g2.brand}\rho_{g2}Generic)$

– patients who have been prescribed generic and product equivalent drugs from different doctors.

$SameDrug(OHIP, name, phone) := \Pi_{OHIP,name,phone}\sigma_{g.brand=p.DIN}(DiffDocs \bowtie_{drug1=g.DIN} \Pi_{g.DIN,g.brand}\rho_{g}Generic \bowtie_{drug2=p.DIN} \Pi_{p.DIN}\rho_{p}Product)$

– patients who have been prescribed any equivalent drugs from different doctors.

$DoctorShopping(OHIP, name, phone) := SameDIN \cup SameBrand \cup SameDrug$

3. Safest ingredient

   Cannot be expressed

4. Drug shortage

   – prescriptions that are unfilled.

   $UnfilledPrescriptions(RxID, drug, patient) :=$
   $\Pi_{RxID,drug,patient}Prescription - \Pi_{RxID,drug,patient}(Prescription \bowtie Filled)$

   – drugs with three or more unfilled prescription.

   $$UnfilledDrugs(drug) := \Pi_{drug}\sigma_{\substack{u1.drug=u2.drug\wedge u2.drug=u3.drug\\ \wedge\\ (u1.patient\neq u2.patient\vee u2.patient\neq u3.patient)\\ \wedge\\ u1.RxID\neq u2.RxID\wedge u2.RxID\neq u3.RxID\\ \wedge\\ u1.RxID\neq u3.RxID}}(\rho_{u1}UnfilledPrescriptions \times \rho_{u2}UnfilledPrescriptions \times \rho_{u3}UnfilledPrescriptions)$$

   – matching drugs with the DIN and manufacturer.
   $DrugShortage(DIN, manufacturer) :=$
   $\Pi_{DIN,manufacturer}(UnfilledDrugs \bowtie_{drug=DIN} (\Pi_{DIN,manufacturer}Generic\cup\Pi_{DIN,manufacturer}Product)$

5. Protecting drug patents

   – all pairs of drugs with active ingredients.

   $AllPairs(din1, name1, din2, name2) := \Pi_{p1.din,p1.name,p2.din,p2.name}\sigma_{p1.din>p2.din}(\rho_{p1}Product\times$

$\rho_{p2}Product)$

– pairs of drugs where drug 1 has an ingredient drug 2 doesn't.

$MissingDIN2(din1, name1, din2, name2) := \Pi_{din1,name1,din2,name2}((AllPairs \bowtie_{din1=DIN} \Pi_{DIN,ingredient}Contains) - (AllPairs \bowtie_{din2=DIN} \Pi_{DIN,ingredient}Contains))$

– pairs of drugs where drug 2 has an ingredient drug 1 doesn't.

$MissingDIN1(din1, name1, din2, name2) := \Pi_{din1,name1,din2,name2}((AllPairs \bowtie_{din2=DIN} \Pi_{DIN,ingredient}Contains) - (AllPairs \bowtie_{din1=DIN} \Pi_{DIN,ingredient}Contains))$

– pairs of drugs with all the same active ingredients.

$DrugPatents(din1, name1, din2, name2) := AllPairs - MissingDIN2 - MissingDIN1$

6. Patients at risk

– all drugs matched to their ingredients.

$AllDrugIngredients(DIN, ingredient) :=$
$\Pi_{DIN,ingredient}(Contains \bowtie Product) \cup (\Pi_{g.DIN,c.ingredient}\rho_c Contains \bowtie_{g.brand=c.DIN} \rho_g Generic)$

– prescriptions matched to the ingredients in their drugs.

$PrescriptionIngredients(RxID, date, patient, doctor, ingredient) :=$
$\Pi_{RxID,date,patient,doctor,ingredient}(AllDrugIngredients \bowtie_{DIN=drug} Prescription)$

– all prescriptions that are on the same date, same doctor and same patient.

$SameDocSamePatSameDay(doctor, date, d1ingredient, d2ingredient) :=$
$\Pi_{p1.doctor,p1.date,p1.ingredient,p2.ingredient}\sigma_{\substack{p1.RxID>p2.RxID \wedge p1.doctor=p2.doctor \\ \wedge \\ p1.patient=p2.patient \wedge p1.date=p2.date}}$
$(PrescriptionIngredients \times PrescriptionIngredients)$

– prescriptions that interact with each other.

$RiskPatients(doctor, date) :=$
$\Pi_{doctor,date}(SameDocSamePatSameDay \bowtie_{d1ingredient=ingredient1 \wedge d2ingredient=ingredient2} Interaction)$

7. Many generics

Cannot be expressed

8. Lots of competition

– all manufacturers of brand drugs.

$Manufacturers(manufacturer) := \Pi_{manufacturer}(Product)$

– manufacturers of at least one brand name without generics of same brand name by the same manufacturer.

$SameManufacturer(DIN, manufacturer) := \Pi_{DIN,manufacturer}(Product) -$
$\Pi_{p.DIN,p.manufacturer}(\rho_p Product \bowtie_{p.DIN=g.brand \wedge p.manufacturer=g.manufacturer} \rho_g Generic)$

– manufacturers of brand names with at least one brand name without generics of same brand name by another manufacturer.

$DifferentManufacturer(DIN, manufacturer) := \Pi_{DIN,manufacturer}(Product) -$
$\Pi_{p.DIN,p.manufacturer}(\rho_p Product \bowtie_{p.DIN=g.brand \wedge p.manufacturer \neq g.manufacturer} \rho_g Generic)$

– manufacturers with every brand name with a generic for the brand name with the same manufacturer and one with a different manufacturer

$Competition(manufacturer) := Manufacturers -$
$\Pi_{manufacturer} SameManufacturer - \Pi_{manufacturer} DifferentManufacturer$

Part 2: Additional Integrity Constraints

1. Symmetry

   $Interaction - \Pi_{ingredient2,ingredient1}(Interaction) = \emptyset$

2. Don't surpass those with seniority

   Cannot be expressed

3. Brand-name first

   – all prescriptions for generic drugs.

   $AllGen(date, gen, brand, doctor) := \Pi_{Prescription.date,Generic.DIN,Generic.brand,Prescription.doctor}($
   $\sigma_{Generic.DIN=Prescription.drug}(Prescription \times Generic))$

   – generic prescriptions where the doctor previously prescribed the brand-name equivalent.

   $ValidGen(date, gen, brand, doctor) := \Pi_{AllGen.date,AllGen.gen,AllGen.brand,AllGen.doctor}($
   $\sigma_{\substack{AllGen.brand=Prescription.drug \\ \wedge \\ AllGen.date>Prescription.date \\ \wedge \\ AllGen.doctor=Prescription.doctor}} (AllGen \times Prescription))$

   – all generic prescriptions must have prior brand-name prescription.

   $AllGen - ValidGen = \emptyset$